Spain: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- Spain is most exposed in upstream gas and upstream oil, where its Energy Security is higher than only 4% and 10% of the other economies scored. Production falls short of consumption in both, and no proved reserves are recorded.[1]
- Its strongest openings are in midstream electric vehicles, downstream solar and upstream solar, where its Economic Opportunity is higher than 99%, 95% and 95% of the other economies scored. Investment, production and trade, respectively, push these scores up.[1]
- On the paper's weighting, the Strategic Index ranks midstream batteries 1st among its own sectors, almost tied with downstream solar. Policy activity agrees for that sector, which combines top priority with the most policy.[1]
- The Partnership Strength Index rates Portugal highest for Spain, as a reliable supplier. China is its 1st trade partner but ranks 45th on Partnership Strength.[1,2]

1. Energy security risks and why

4 of Spain's 30 placed sectors, those whose scores can both be compared with other economies, sit in the two low-energy-security corners. Energy Security measures how well a country can rely on what it needs in a sector. Measured against other economies, its lowest Energy Security is in upstream gas (0.25), upstream oil (0.36) and downstream coal (0.42). These are higher than only 4%, 10% and 23% of the other 97, 96 and 98 economies scored in these sectors. Both upstream sectors sit in the Upgrade corner; downstream coal and downstream gas sit in the Secure corner. The paper's framework associates the Upgrade corner with building domestic capability and the Secure corner with securing supply through trusted partners.[1]

Spain's high-priority sectors, those in the Upgrade corner ranked by the Strategic Index, are upstream gas and upstream oil. In both, the score is low mainly because of energy imports and reserves. Gas production minus consumption was -1.13 EJ in 2025, and for oil it was -2.65 EJ. Both figures take production from the U.S. Energy Information Administration's international statistics, as the Energy Institute does not report it separately. No proved gas or oil reserves are recorded, which the index counts as zero; filled-in data carry 28.6% of the weight of each score.[1,3,4]

For context, its largest suppliers of upstream gas goods in 2024 were Algeria (41.3% of imports, which totalled USD 9.59 bn), Russia (22.0%) and the United States (21.2%). For upstream oil, imports worth USD 45.7 bn came from the United States (16.4%), Nigeria (10.6%), Brazil (10.2%) and others.[2]

In downstream coal, energy access pulls the score down. Coal consumption was 1.81 GJ per person in 2025, and consumption per person changed by -30.2% over the years to 2025.[3]

2. Economic opportunities and why

27 of the 30 placed sectors sit in the two high-opportunity corners, where Economic Opportunity is above the sector's middle economy. Economic Opportunity measures how well placed a country is to create value in a sector. The highest Economic Opportunity scores are in midstream electric vehicles, downstream solar and upstream solar, all in the Grow corner. They are higher than 99%, 95% and 95% of the other 96, 147 and 92 economies scored. The paper's framework associates the Grow corner with export promotion.[1]

In midstream electric vehicles, investment and foreign dependency push the score up. On growth in investment Spain scores higher than 100% of the other economies, and on annual investment higher than 97%. For foreign dependency the index uses the EU's figure, carried by each EU member, not Spain's own: electric-vehicle production of 2.37 million vehicles in 2024, 13.7% of the world's.[1,5]

In downstream solar, production pushes the score up: solar generation was 64.5 TWh in 2025. In upstream solar, trade does: on export feasibility, which reads how related its existing exports are to these goods, Spain scores 0.45, higher than 98% of the other economies.[1,3,6]

For context, France took 23.5% of Spain's midstream electric vehicles exports of USD 22.6 bn in 2024, Germany 11.7% and the United Kingdom 11.6%.[2]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks a country's own sectors by combining opportunity, security need, policy activity, the stage and the technology's low-emission value. On the paper's weighting, it puts midstream batteries 1st (0.64), downstream solar 2nd, upstream batteries 3rd, upstream wind 4th and upstream solar 5th among Spain's own sectors, all in the Grow corner. They are almost tied, with gaps of 0.000, 0.001, 0.002, 0.003 and 0.009 to the next sector.[1]

The allied pick, its leading upstream or midstream sector once sectors without a recorded resource base are set aside, is midstream batteries, 0.020 ahead of upstream nuclear. The runner-up comes from further down because upstream batteries, upstream wind, upstream solar and upstream electric grid are screened out: no resource base is recorded for them.[7]

Policy activity counts International Energy Agency Policies and Measures records and the Climate Action Tracker rating, read against each technology. Priority here means the Strategic Index without its policy term. The sector with top priority and the most policy is midstream batteries, while upstream electric vehicles combines top priority with the least policy. The reverse holds for downstream green hydrogen and downstream electric vehicles, which combine bottom priority with the most policy. Spain carries the European Union's Climate Action Tracker rating, Insufficient (November 2024), and has 362 Policies and Measures records, 200 of which name a technology. For context, Country Insights also lists industrial-policy measures recorded by Global Trade Alert, some of them EU-wide; policy activity does not count them.[1,8,9]

4. Partnership priorities and why

The Partnership Strength Index rates partners as suppliers (Friendshore), as markets for exports (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. From Spain's viewpoint it ranks Portugal 1st, France 2nd, the Netherlands 3rd, the United States 4th and Italy 5th of 191 partners. Portugal, France and the United States stand out most as a reliable supplier, the Netherlands and Italy as an export market. As a supplier, the United States scores best in upstream gas, upstream oil,

downstream gas, midstream wind and downstream geothermal. As a market, the Netherlands scores best in upstream solar, upstream nuclear, upstream wind, upstream batteries and upstream electric vehicles.[1]

These priorities largely match today's ties. France is also Spain's 2nd trade partner on average across its energy-goods baskets, the Netherlands its 5th, Italy its 6th, the United States its 7th and Portugal its 9th. Each is a priority partner with strong ties today, and UN votes put each in the same bloc as Spain. By contrast, China is Spain's 1st trade partner, with 12.2% of its imports on average, but ranks 45th on Partnership Strength, where it stands out most as a development partner. UN votes put Spain and China in opposite blocs.[1,2,10]

As a partner to the coalition, Spain ranks 7th of 192 economies, on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads upstream and midstream production across coalition members, its largest designed share is 1.9% of midstream green hydrogen.[1,7]

5. Data notes and caveats

Baseline of 12 October 2026; details in Country Insights.

- Scores compare Spain with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Many allied picks are close calls: in the paper, 37% of them are within 0.01 of the runner-up.
- In the allied network design the European Union is one producer; Spain's shares are derived from its weight within the EU.
- The Climate Action Tracker rates the EU as a whole; Spain carries the EU's rating.
- Spain's counts leave out the EU's own 438 International Energy Agency policy records.
- Trade figures count trade with other EU members.
- Each EU member carries the EU's announced 2035 manufacturing-capacity shares and electric-vehicle figures.

Sources

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